FortyGuard Products Evaluation Report

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Scope: Temperature Dashboard(R), Temperature API(R), API documentation, and bounded live API probes. API key and account-specific sensitive values are redacted from evidence.

Executive Summary

FortyGuard has a compelling product direction: hyperlocal temperature intelligence, asynchronous geospatial processing, and a dashboard surface that immediately communicates the map-based climate-tech use case. The strongest design choice is the async activity_id pattern, which is appropriate for compute-heavy heatmap, satellite, street-view, and report workflows.

The main gaps are reliability and operational developer experience. A valid POST /v1/heatmap request returned 500 Internal Server Error in live testing, and an invalid unclosed polygon also returned 500 instead of a validation response. The API exposes a useful OpenAPI 3.1 schema, but it is not surfaced as an integrated interactive documentation path.

Methodology

Area	What was checked
Live API probe	Heatmap submission, geometry validation, status lookup, env_params schema, credits usage, satellite short polling, headers.
Docs review	Quickstart, endpoint pages, credits usage, known limitations, release notes, and hidden OpenAPI/ReDoc assets.
Visual audit	Docs screenshots plus dashboard /map at desktop and mobile widths. Fresh /map sessions redirected to /login.

Strengths

- The async activity_id submission and status polling model is the right architecture for heavy geospatial workloads.

- The credit-on-success model described in the docs is developer-friendly and builds trust.

- The public OpenAPI 3.1 schema at /openapi.json is valuable for Postman import, SDK generation, and automated testing.

- The dashboard login screen is polished, responsive, and visually aligned with the temperature-map product category.

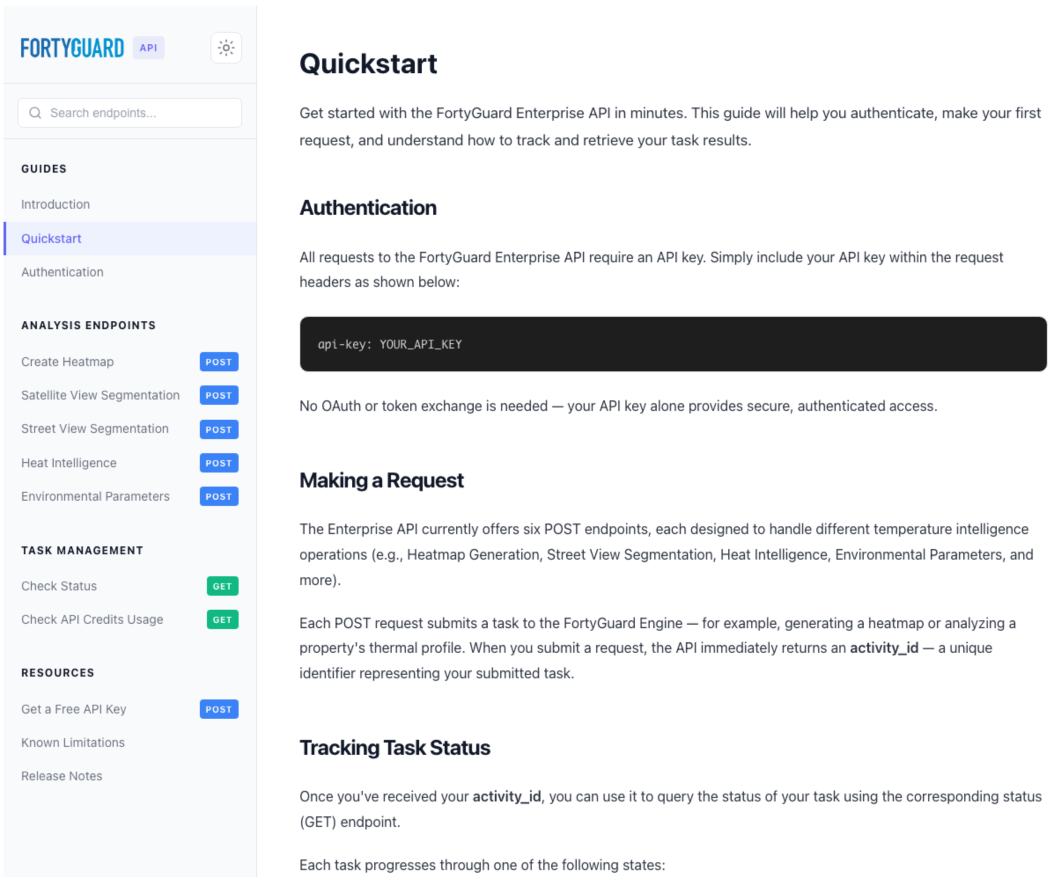
Key Findings

Priority	Finding	Evidence	Recommendation
P0	Heatmap happy path fails	Valid POST /v1/heatmap returned 500.	Fix as release blocker and add regression coverage for minimal valid polygons.
P1	Geometry validation escapes as server error	Unclosed polygon returned 500.	Validate GeoJSON before enqueueing and return 400/422 with a specific error code.
P1	Unknown activity id returns 403	GET /v1/status/not-a-real-id returned 403.	Return 404 for missing resources; reserve 403 for authenticated-but-forbidden access.
P1	No operational headers	No X-RateLimit, Retry-After, X-Credits-Remaining, or request-id header observed.	Add quota, credit, retry, and correlation headers to authenticated responses.

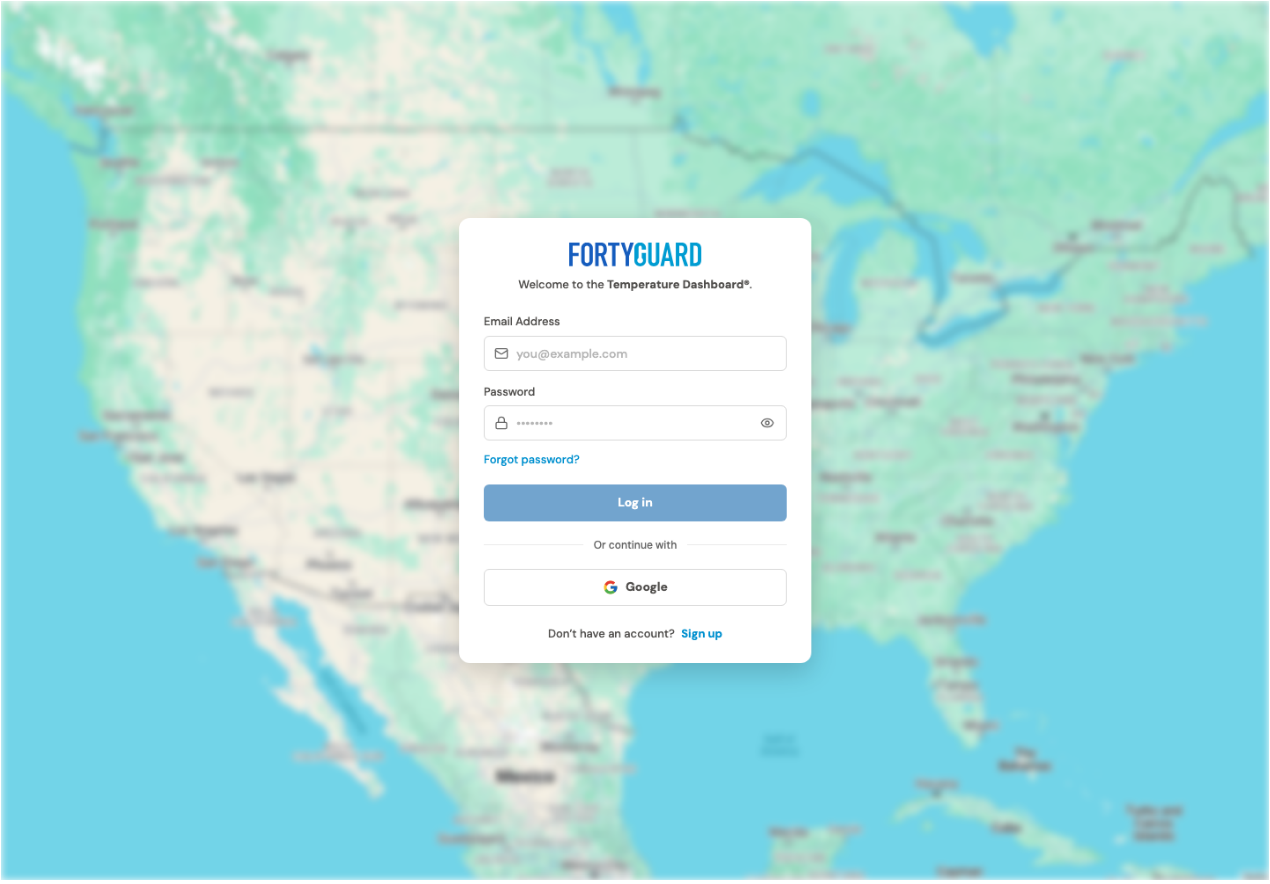
Priority	Finding	Evidence	Recommendation
P2	Credits docs mismatch	Docs label credits usage as GET form; OpenAPI shows POST /v1/system/fetch-api-key-usage.	Document raw HTTP request/response schema and align method labels.
P2	Env params customization unclear	Docs mention customizable params; OpenAPI request has no parameters selector.	Expose selector in schema or update docs wording.
P2	Satellite schema typo	Docs/result use orignal_image.	Support original_image and deprecate the typo.

Visual Evidence

Fresh browser sessions could render the public documentation, but direct dashboard /map access redirected to /login at desktop and mobile sizes. This means the login/access gate was verified visually, while authenticated map workflows require dashboard credentials or an existing session for a fair review.



Docs quickstart: clean navigation and readable API onboarding.



Dashboard /map in a fresh session: redirected to the polished login experience.

Recommended Priorities

1. Fix reliability first

Repair the heatmap endpoint, add geometry validation, and introduce happy-path plus negative-path regression tests.

2. Standardize API errors

Use one error envelope across custom auth errors and FastAPI validation errors, with code, message, doc_url, and request_id.

3. Add operational telemetry

Return credit, rate-limit, retry, and correlation headers so developers can operate safely without extra quota calls.

4. Improve async workflow ergonomics

Add webhook callbacks, task cancellation, idempotency keys, and documented polling/backoff behavior.

5. Tighten documentation accuracy

Expose OpenAPI in the main docs, reconcile method labels, clarify env_params customization, fix field names, and document concrete rate limits.

Suggested Form Ratings

Area	Rating	Reason
Dashboard usability	6/10	Polished login and brand presentation, but authenticated map workflow could not be freshly verified and needs clearer onboarding.

Area	Rating	Reason
API	5/10	Good async foundation, but primary heatmap happy path returning 500 is a serious reliability issue.
Overall	6/10	Strong product potential, but reliability, docs accuracy, and operational DX need tightening.

Final Takeaway

FortyGuard feels like a technically ambitious platform with a real climate-tech use case. The fastest path to enterprise-ready trust is to stabilize the heatmap happy path, validate geospatial inputs cleanly, make quota/credit state visible in every response, and turn the hidden OpenAPI schema into a first-class developer experience.

Evidence Files

- evidence/api_probe_2026-06-24.jsonl

- evidence/visual_audit_2026-06-24.json

- evidence/docs_pages_2026-06-24.json

- evidence/limitations_full_text_2026-06-24.txt